Answer **one** question from this section.

**9** **(a)** Distinguish between *velocity* and *angular velocity* for a body undergoing oscillations.

.....

.....

.....

..... [2]

**(b)** A toy pendulum consists of a sphere of mass 0.800 kg, attached firmly to one end of a rod of negligible mass. The other end of the rod is pivoted freely at point O. The distance between the centre of gravity of the toy to O is 0.600 m. The structure is held in a position such that the rod is at an angle of 6.00° from the vertical, as shown in Fig. 9.1.

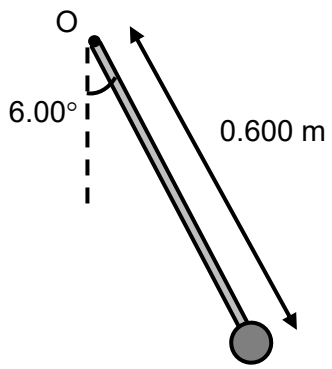


**Fig. 9.1**

The structure is then released from rest and oscillates in simple harmonic motion. At one instant during the oscillation, the toy is directly below O, as shown in Fig. 9.2.

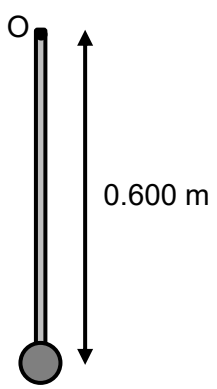


**Fig. 9.2**

**(i)** On Fig. 9.2, indicate the forces acting on the sphere. [2]

**(ii)** Show that the linear speed of the sphere at the instant in Fig. 9.2 is 0.254 m $s^{-1}$.

[2]

**(iii)** Determine the force *F* exerted on the pivot by the rod in Fig. 9.2.

*F* = .....N [2]

**(iv)** On Fig. 9.3, sketch the velocity-time graph of the sphere with correct labels assuming that the sphere is at maximum displacement at *t* = 0 s. Label the period as *T*.

**Fig. 9.3**

[2]

**(c) (i)** State what is meant by *resonance*.

.....

.....

..... [2]

**(ii)** A ship of mass 1020 kg undergoes simple harmonic motion in the vertical direction due to surface water waves that are incident on the ship. The water waves of amplitude of 0.18 m and frequency 0.56 Hz cause resonance in the vertical motion of the ship.

**1.** Write an expression for the vertical displacement $x$ as a function of the time $t$ associated with the motion of the ship.

[2]

**2.** Calculate the magnitude of acceleration of the sailing boat when its displacement from its equilibrium position is 0.080 m.

$a$ = ..... m $s^{-2}$ [2]

**3.** Calculate the maximum kinetic energy of the ship during the vertical oscillation.

maximum kinetic energy = .....J [2]

**4.** On Fig. 9.4, sketch a labelled graph showing the variation of kinetic energy of the ship with time.

**Fig. 9.4**

[2]